Définition .1.

Let X be a complex manifold and

Proposition .1.

Let X be a complex manifold and let r be a positive integer.
The categories $\mathrm{VectBun}^r(X)$ and $\mathrm{LocFree}_r(X)$ are equivalent.

PROOF

We define

$$\begin{array}{rccl} \mathcal{F}: & \mathrm{VectBun}_r(X) & \longrightarrow & \mathrm{LocFree}_r(X) \\ & E & \longmapsto & \mathcal{F}_E \end{array},$$

where $\mathcal{F}_E$ is the sheaf of sections of E. It is well defined by ??.
It is a functor : given a morphism of vector bundles $f : E_1 \to E_2$, we define for each open set U of X the function

$$\begin{array}{rccl} \varphi_U: & \mathcal{F}_1(U) & \longrightarrow & \mathcal{F}_2(U) \\ & s & \longmapsto & f \circ s \end{array}.$$

φ clearly defines a morphism of sheaves.
In order to construct a quasi-inverse, let $\mathcal{F} \in \mathcal{S}_X$ and $\psi : \mathcal{F}(U_i) \to {\mathscr{O}_{U_i}}^{\oplus r}$ local trivialisations on an open cover $\{U_i, i \in I\}$ of X.

□

Définition .2.

Let $\mathcal{E}$ be a locally free $\mathscr{O}_X$-modules.
A *connection* is a $\mathbb{C}$-linear application $\nabla : \mathcal{E} \to \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^1_X$ satisfying the *Liebniz relation* : for any local sections s and local function f, then

$$\nabla(fs) = df \otimes s + f\nabla(s).$$

A connection ∇ is said to be *flat* (or *integrable*) if the induced

$$\begin{array}{rccl} \nabla': & \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^1_X & \longrightarrow & \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^2_X \\ & s \otimes \omega & \longmapsto & \nabla s \wedge \omega + s \otimes dw \end{array}$$

(where d is as usual defined by $d(g \cdot dz) = dg \wedge dz$) satisfies $\nabla' \circ \nabla = 0$.

Définition .3.

We define the category $\mathrm{LocFree}^{\nabla}(X)$ with

— locally free $\mathscr{O}_X$-modules equipped with flat connections as objects,

— morphisms of sheaves $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ making

$$\begin{tikzcd}
\mathcal{E}_1 \arrow[r, "\nabla_1"] \arrow[d, "\varphi"] & \mathcal{E}_1 \otimes_{\mathscr{O}_X} \Omega^1_X \arrow[d, "\varphi\otimes 1"] \\
\mathcal{E}_2 \arrow[r, "\nabla_2"] & \mathcal{E}_2 \otimes_{\mathscr{O}_X} \Omega^1_X
\end{tikzcd}$$

commutes as morphisms between $(\mathcal{E}_1, \nabla_1)$ and $(\mathcal{E}_2, \nabla_2)$.

Théorème .2 (Riemann-Hilbert correspondence, complex manifold case).

The fonctor

$$\begin{array}{rcl} F: & \mathrm{LocFree}^{\nabla}(X) & \longrightarrow \mathrm{LocSys}(X) \\ & (\mathcal{E}, \nabla) & \longmapsto \mathcal{E}^{\nabla} := \mathrm{Ker}(\nabla) \end{array}$$

induces an equivalence of categories.
It's inverse is given by

$$\begin{array}{rcl} G: & \mathrm{LocSys}(X) & \longrightarrow \mathrm{LocFree}^{\nabla}(X) \\ & \mathcal{A} & \longmapsto (\mathscr{O}_X \otimes_{\mathbb{C}} \mathcal{A}, d \otimes 1) \end{array}$$

Proof

$\square$